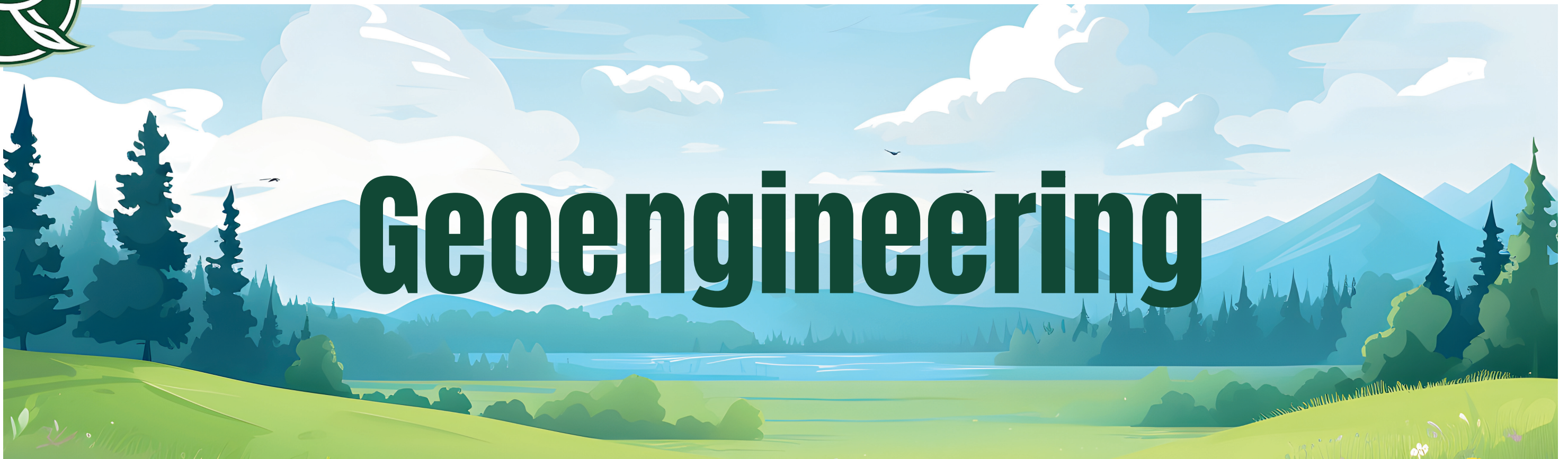




Adapted from Russell LM et al. 2012, "Ecosystem impacts of geoengineering." See last page for full citation.

Introduction

Geoengineering is intentionally changing Earth's features, in this case to reduce **climate change**. This study focuses on the need for research which identifies and quantifies the potential consequences of geoengineering.

These consequences are part of a broader series of questions (governance, ethics, social and physical science) to address before beginning geoengineering. The proposed geoengineering methods covered are **Carbon Dioxide Removal (CDR)** and **Solar Radiation Management (SRM)**.



The Climate System

The Earth has to balance incoming energy from the Sun¹ and outgoing heat radiation, both of which control the Earth's temperature and climate. **Albedo** is the fraction of this incoming energy that is reflected back into space by clouds, **aerosols**, icecaps, and bright surfaces. The other energy is absorbed into the Earth and its atmosphere, heating them up. **Greenhouse gases** (GHG) in the atmosphere reduce the amount of energy that escapes to space by about 60%, making Earth's atmosphere trap heat similarly to the walls of a greenhouse: letting energy in, but not back out. This is known as the greenhouse effect, which drives **global warming** (climate change).

¹Including ultraviolet (UV), visible, and infrared (IR) light



Ecosystems

The world's ecosystems are incredibly important to humanity through their impacts on economy, climate, disease prevention, and waste, as well as recreational and cultural activities. An ecosystem is made up of its own unique combination of **flora** and **fauna**, often referred to as its biodiversity.



Climate change destroys ecosystems, creating a wide range of impacts on humans. However, CDR and SRM methods used to reduce climate change are risky, as there is no way to know their ecological effects until implemented. Unseen side effects of CDR and SRM methods could worsen the loss of biodiversity, making research necessary before geoengineering methods can be put in place.



Governance of CDR & SRM Research

Though past efforts have provided suggestions to governments, they rely on national governments alone, even though international systems are favorable. Most of the research community believes government involvement is necessary for potentially risky geoengineering tests. In particular, we need responsible, cooperative research that emphasizes human and environmental benefits, along with public involvement.



“Control” Scenario

As GHG levels rise and cause increasingly severe impacts on the Earth, it is impossible to know what ecosystems will look like in the future. Therefore, in order to compare the impacts of a future with a geoengineering method to one without, researchers measure them against a “control,” a hypothetical Earth with increased GHG levels and proportionally increased impacts to what we see now.

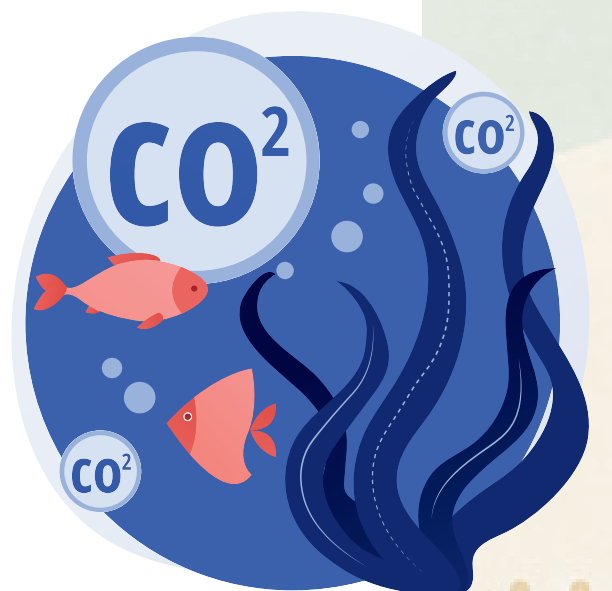


Carbon Dioxide Removal Methods (CDR)

Getting CDR large enough to have significant effects would be slow (likely decades), but with long lasting effects. Only CDR would address the CO₂ concentrations responsible for both climate change and other effects like ocean acidification, because its aim is to remove CO₂ that is currently in the air.

Ocean Acidification

Ocean acidification is one of the many effects of rising atmospheric CO₂: the CO₂ dissolves into ocean water, increasing its level of acidity. This kills off marine biodiversity, especially in coral reefs. As CDR methods reduce the amount of CO₂ in the atmosphere, ocean acidification would also be reduced. On the other hand, SRM simply reflects light, assisting in other areas but leaving CO₂-specific effects like ocean acidification to continue rising unchecked.



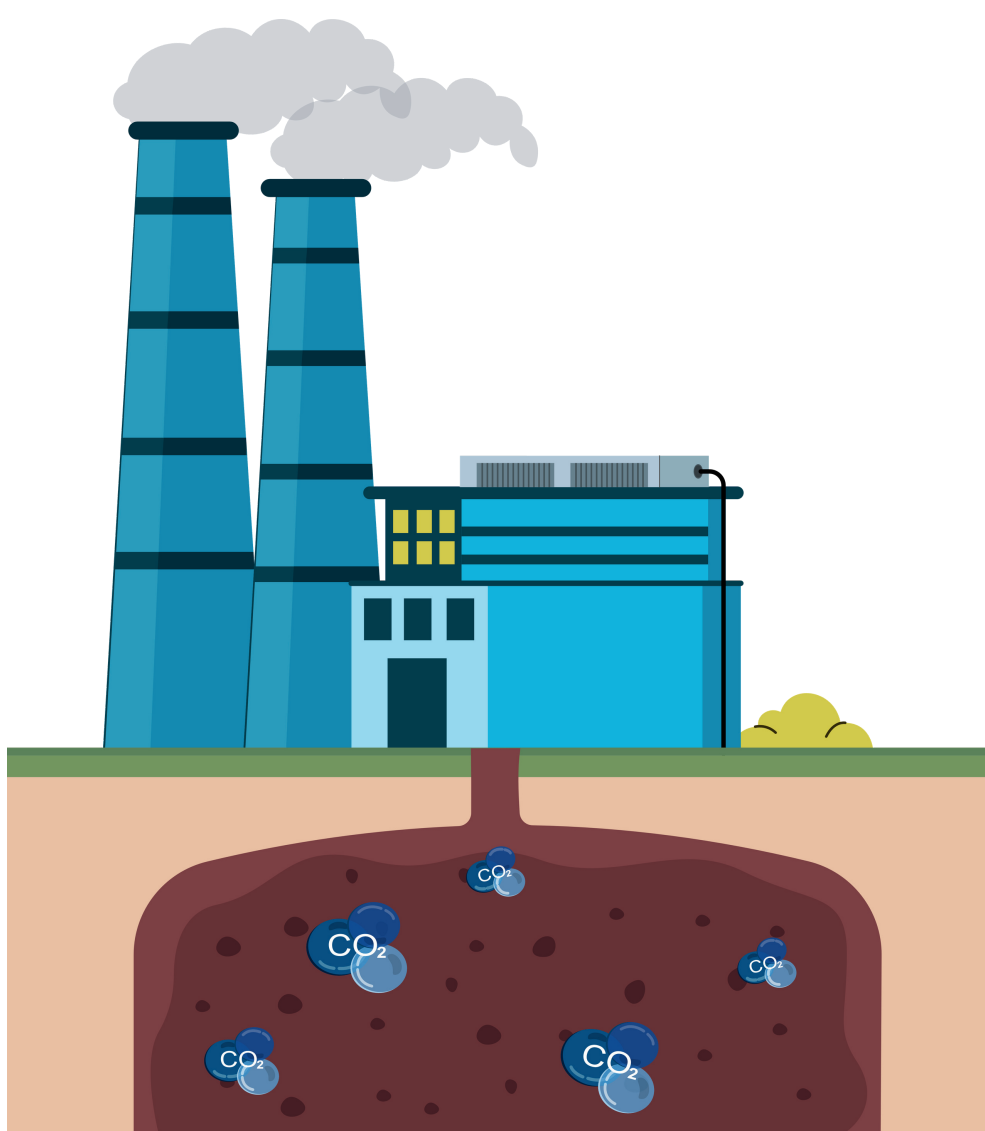


Afforestation

Afforestation is the process of planting more trees to create natural **carbon sinks**. As trees grow, they naturally “breathe” in carbon dioxide, keeping it out of the atmosphere until they decay. Different forests have different rates at which they grow and take in carbon, making some areas like tropical forests better options for afforestation. However, afforestation would have to be implemented on an extremely large scale for it to have a tangible impact on climate change. Afforestation is only possible on abandoned land, and would likely use lots of water and nitrogen, potentially having mixed impacts on the surrounding areas. These forests must also be regularly maintained.



Engineered Carbon Capture + Storage



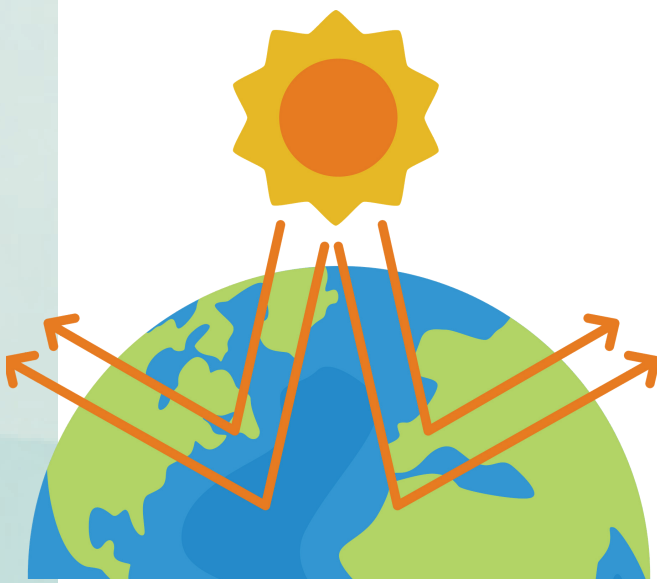
Engineered carbon capture refers to the use of chemicals to absorb CO₂ out of the atmosphere. It can also refer to the use of strong bases (the opposite of acids) to neutralize CO₂. After capture, CO₂ would be stored in human-made facilities, likely underground, creating minimal, or at least localized, environmental impacts. However, resources used for carbon capture may take a toll on the environment.

Ocean Fertilization



skabsorb approximately 40% of captured CO₂ per year. Fertilizing oceans would increase the amount of phytoplankton, which would increase the amount of CO₂ the phytoplankton can absorb. Three main restrictions of phytoplankton growth are the amount of iron, phosphate, and nitrate in surface waters, where phytoplankton live. Adding these nutrients could be done by moving nutrient-heavy deep-sea waters towards the surface, or by using man-made fertilizers.

Solar Radiation Management (SRM)



Stratospheric Aerosol Injection

Stratospheric aerosol injection is the process of pumping particles into the **stratosphere** to increase albedo (shininess). This reduces the energy that reaches the Earth's surface, mimicking the cooling period created by volcanic explosions. The main aerosol used for injection would be sulfate. Drawbacks may include acid rain (which contributes to ocean acidification) and reduction of **UVB** light. Overall, it is hard to predict the side effects of creating such a large scale and long lasting artificial explosion.





Cloud Albedo Enhancement

Cloud albedo enhancement essentially makes clouds shinier by using aerosols which water vapor can condense on, in effect, reflecting light before it hits the sea. However, cutting sunlight levels reaching the Earth could have negative impacts on both localized marine ecosystems and medium scale atmospheric systems, changing weather patterns like El Niño and the West African Monsoon, which are vital to nature and human life in large parts of the world.

SRM's Problems

SRM can potentially change the world in a combination of ways historically unknown by species currently living on the planet, meaning it is hard to predict the ecological consequences it may have. Additionally, SRM methods must be renewed repeatedly, or else the Earth will warm rapidly, affecting weather and almost certainly causing impacts worse than current global warming (drought, fire risk, famine, etc.). If SRM is implemented, researchers still need to pay attention to the amounts of CO₂ in the atmosphere and its impact on ocean acidification.



Uncertainties

Detecting and quantifying the impact of geoengineering activities on **u** is difficult. Research should distinguish natural versus man-made changes to the climate.

Preliminary Research Plan

Framing the Question

It is important to design the locations and durations of CDR and SRM studies so that they capture the responses of ecosystems and their time scales, especially with large uncertainties. There is also the issue of determining a control to compare results to, as well as observing effects on land, sea, and air simultaneously.

Numerical Models + Experiments

Numerical models are created on computers, using data that we already have along with variables to predict (model) potential outcomes.

Analog Experiments

Unlike numerical models, **analog experiments** are done physically in labs. They can mimic geoengineering methods while saving time and space, useful for developing hypotheses to test with larger simulations and experiments.



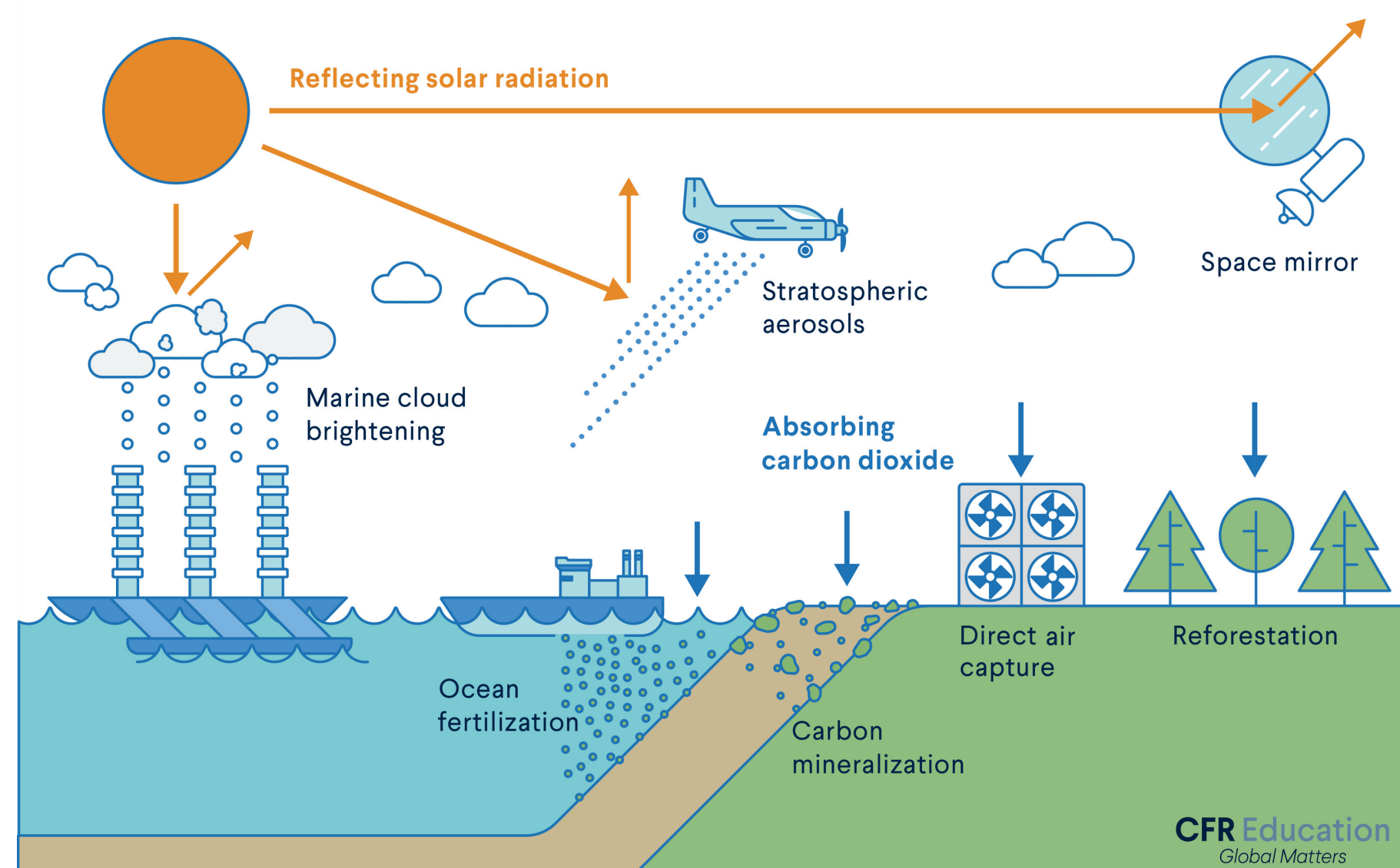
Field Experiments

Field experiments using direct manipulation should be preceded by **risk assessments** and numerical modeling. Optimal experiments need physical, chemical, optical, and biological measurements, designed by experts working together. These experiments can be small: for example, some SRM studies can be performed without aerosols or cloud seeding, as clear plastic sheets can increase the amount of diffuse light reaching a forest.

Conclusions

It is important to research before large-scale implementation of geoengineering, as geoengineering will create conditions different from the past, present, and a future without it. Additionally, research can optimize implementation by identifying the most reliable and effective methods and locations. Research may find that geoengineering will create unacceptable conditions and changes, so supporting research should not be seen as advocating for geoengineering, but supporting the basis for the best and most scientific policy decisions.

Potential Geoengineering Solutions to Fight Climate Change



Recommendations

Each geoengineering method should be thoroughly researched and perhaps implemented on small scales to see the ecological impacts. Experiments should not be the only action taken, as research into ecosystems and other Earth systems must also be carried out in case there are better solutions we have yet to discover.

Glossary

Aerosol — a substance enclosed under pressure and able to be released as a fine spray, typically by means of a propellant gas

Analog Experiments — experiments in artificial settings

Carbon Sinks — natural or man made areas that store carbon and keep it out of the atmosphere

Climate Change — the overall change in weather patterns over a long period of time; in particular, global warming and other drastic human-caused changes

Ecology — the study of living organisms interactions with each other and the environment

Fauna — animals

Field experiments — experiments using some controlled elements of traditional lab experiments, but in natural, real-world settings

Flora — plants

Global warming — heating of earth, frequently used to refer to the man-made conditions that are (and have been) creating climate change in the modern era

Numerical modeling — the use of computers to predict the outcome of complex systems

Phytoplankton — bacteria and protists that photosynthesize in the well-lit surface areas of bodies of water, forming the base of marine and freshwater food webs

Risk assessments — an evaluation analyzing things that could go wrong

Stratosphere — the portion of the Earth's atmosphere around 50 km above the ground with cold temperatures and minimal clouds

UVB — Ultraviolet B light, a type of UV light

Full Citation

Russell LM, Rasch PJ, Mace GM, Jackson RB, Shepherd J, Liss P, Leinen M, Schimel D, Vaughan NE, Janetos AC, Boyd PW, Norby RJ, Caldeira K, Merikanto J, Artaxo P, Melillo J, Morgan MG. Ecosystem impacts of geoengineering: a review for developing a science plan. *Ambio*. 2012 Jun; 41(4):350-69.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3393062/#Sec7>